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Application of a water infiltration model for simulating water repellency of humus soil

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Abstract

One-dimensional infiltration experiments were conducted using hydrophilic and water-repellent soils from the Guishui River Basin to study the effects of soil water repellency on cumulative infiltration (CI) and the infiltration rate (IR). The test results show that, for the hydrophilic soil (HS) sample, CI increases monotonously with time and IR decreases monotonously. For the water-repellent soil (W-RS), however, the following characteristics were observed: (a) There is an inflection point in CI and a sudden increase in IR. Larger values of the initial soil water content produce an earlier and more significant inflection point in CI, and a larger peak value of IR. (b) The post-peak stable IR is greater than the pre-peak value, ignoring the beginning of rapid infiltration, and the overall IR presents a single peak. The applicability of various water infiltration models was analysed for the two soil types. Numerical analysis suggests the following conclusions: (a) For both HS and W-RS, the Kostiakov function, Gamma function, and Beta function (BF) models exhibit good applicability. (b) For W-RS, the Gauss function model not only reflects the monotonous decrease in IR, but also produces a steady IR in the initial stage, a gradual increase before the peak value, and a gradual decrease after the peak value. Similarly, the BF model reflects the monotonous decrease in IR. A piecewise BF reproduces the U-shaped change in rapid infiltration before the inflection point, as well as the gradual increase and right-skewed distribution curve of W-RS infiltration before and after the inflection point. The BF model achieves the best simulation accuracy and has the widest applicability.

KEYWORDS

disappearance of water repellency, plant root growth, probability density function, single peak, soil volume expansion, soil water repellency, temperature rises, water infiltration model

1 | INTRODUCTION

Soil water repellency (WR) is a physical phenomenon whereby water is unable to wet the soil surface of mineral particles (Ritsema & Dekker, 1994; Ritsema, Dekker, Ritsema, & Dekker, 2003). This phenomenon is widely distributed across the world (Zavala, García-Moreno, Gordillo-Rivero, Jordán, & Mataix-Solera, 2014), with higher WR typically occurring in coarse-textured forest soils (Benito, Varela, & Rodríguez-Alleres, 2019). WR changes the three-dimensional distribution and dynamic characteristics of soil water content, resulting in an

uneven distribution of soil moisture (Badía-Villas, González-Pérez, Aznar, Arjona-Gracia, & Martí-Dalmau, 2014; Page-Dumroese, Robichaud, Brown, & Tirocke, 2015). WR not only affects the soil wetability (Sepehrnia, Hajabbasi, Afyuni, & Lichner, 2017), but also impacts other ecological and hydrological processes, for example, restraining water infiltration (Sepehrnia et al., 2017), promoting surface runoff (Amiri, Emami, Mosaddeghi, & Astaraei, 2017; Neris, Tejedor, Rodríguez, Fuentes, & Jiménez, 2013), water and soil erosion (Cawson, Nyman, Smith, Lane, & Sheridan, 2016; Fox, Darboux, & Carrega, 2007), wind erosion, accelerating fertilizer loss (Müller et al., 2018) by influencing

preferential flow (Oostindie, Dekker, Wesseling, & Ritsema, 2008; Rye & Smettem, 2017), and increasing soil carbon content (Muñoz-Rojas et al., 2018), which may lead to a reduction in crop growth (Li, Yao, Tang, Chau, & Feng, 2019).

The hydraulic properties of water-repellent soil (W-RS) are quite different from those of hydrophilic soils (HS). Ward, Roper, Jongepier, and Micin (2015) concluded that, in W-RS, tillage destroys the existing water entry pathways, and slows the infiltration of water into the soil. Doerr et al. (2003) found that WR can reduce the soil infiltration rate (IR) by 10–40%, while DeBano (2000) showed that the horizontal infiltration of W-RS was 25 times slower than that of HS. Keizer et al. (2005) reported that soil IR can be decreased by reducing the potential gradient of the soil matrix. Many studies show that WR is caused by soil particles and aggregates being coated with hydrophobic materials, which can originate from plant litter and residues, microbes, organic fertilizers, or the application of wastewater and artificial hydrophobic agents (Leelamanie & Karube, 2009; Subedi et al., 2012). The influence of these factors on the degree of WR reflects the relationship between WR and the contact angle (CA), which enables measurement of the time dependency of CA (Leelamanie & Karube, 2009). The range of CA attained by wettable soils is less than 90° , whereas W-RS have $CA > 90^\circ$ (Lourenço et al., 2018). Severe hydrophobicity occurs when the CA is greater than 90° . In this case, water does not spontaneously infiltrate. Subcritical repellency occurs when the contact angle is less than 90° , whereby the soil wets spontaneously but with a reduced IR (Tillman, Scotter, Wallis, & Clothier, 1989). For the same CA, the matrix suction gradually decreases as the volume of the liquid bridge increases (Graber, Tagger, & Wallach, 2009; Hamlett et al., 2011). When the volume of the liquid bridge reaches a certain level, the matrix suction becomes negative. The WR effect of granules then begins to occur (Subedi et al., 2012; 2013). The CA of hydrophobic media normally decreases with continuous contact with water, eventually allowing water imbibition (Subedi et al., 2013). Arye, Nadav, and Chen (2007) investigated the main imbibition relationship between water saturation and capillary pressure using the capillary rise test, and found that organic matter is likely to detach from the soil particles and be dissolved into the soil solution. This, in turn, decreases the equilibrium CA. The effect of WR on infiltration is very complex because of the unstable wetting fronts, which result in finger-pattern preferential flow paths (Rye & Smettem, 2017; Wang et al., 2000) and hysteresis in soil water retention (Arye et al., 2007). However, WR is a dynamic property (that generally decreases as the soil wets) and, therefore, IR is affected during the process itself, resulting in IR curves that do not correspond to the traditional infiltration theory.

The regulation of surface runoff and infiltration is an important manifestation of the ecological hydrological function of WR (White, Lockington, & Gibbes, 2017). Because WR can create unstable water flow within the soil matrix (De Jonge, Jacobsen, & Moldrup, 1999), the process of water infiltration is relatively complex. Current understanding of the infiltration process in W-RS is limited to the fact that WR can reduce soil IR (Xiao, Sun, Hu, & Kidron, 2019). In fact, as water infiltration continues, the IR does not decrease monotonously

and, contrary to infiltration in wettable soils, can increase with time (Ren, Zhao, Wang, Gong, & Tian, 2018; Wang et al., 2000). This phenomenon generates infiltration curves with a double slope (transient infiltration curve followed by a steady-state section) (Vogelmann, Reichert, Prevedello, Awe, & Cerdà, 2017). For forest soils with strong WR, water infiltration is not stable (Rye & Smettem, 2017), nor is it strictly in accordance with the three-stage process of HS infiltration. Instead, with continuous water infiltration, the WR gradually disappears and the IR appears to mutate (Burch, Moore, & Burns, 1989; Diehl, 2013). The effect of WR is very evident in cumulative infiltration (CI), which exhibits a double-slope curve (Vogelmann et al., 2017). Inaccurate fitting indicates that the Haverkamp model (Haverkamp, Ross, Smettem, & Parlange, 1994) should not be applied to such curves.

Leighton-Boyce, Doerr, Shakesby, and Walsh (2007) and Pierson et al. (2008) found that, under WR conditions, the IR slope gradually increases with continuous rainfall. Doerr, Shakesby, and Walsh (2000) showed that, during the whole rainfall process, the IR of W-RS first decreases, then increases, and then decreases to the lowest value and remains stable. This phenomenon is particularly obvious in forest soils (Neris et al., 2013; Ritsema et al., 2003). Filipović et al. (2018) used HYDRUS (2D/3D) to invert the hydraulic properties of W-RS under drought conditions. They found that the CI of W-RS exhibits a non-smooth, step-like growth trend, whereas the IR first decreases and then increases. Rye and Smettem (2017) believe that including only WR in the model enables a correct assessment of the hydrological process. Müller et al. (2018) considered WR to be an important factor in any hydrological model.

At present, the phenomenon of increasing IR is being ignored and water infiltration is generalized as a monotonously decreasing process. Traditional infiltration models (e.g., those of Green–Ampt, Philip, Kostiakov, and Horton) and piecewise function models are still used to fit the double-slope infiltration process (De Almeida et al., 2018; Moret-Fernández et al., 2019) in the model developed by Haverkamp et al. (1994), and the corresponding infiltration curve indicates that the traditional model should not be applied to this kind of curve. A piecewise Kostiakov function (PKF) has been used to calculate the IR, resulting in a discontinuity at the inflection point (Ren et al., 2018). This contradicts the physical phenomenon whereby WR fades away and only one maximum IR exists at the inflection point. Ren et al. (2018) used a Gauss function (GF) and a piecewise Gauss function (PGF) to fit the IR of W-RS. Although the GF reflects the process of the increasing and then decreasing IR in W-RS, it is difficult to describe the gradual decrease in IR after the infiltration starts.

In this study, water infiltration is analysed in two types of soil using the Kostiakov function (KF), PKF, GF, PGF, Fourier series function (FSF), Gamma function (GMF), Beta function (BF), and piecewise Beta function (PBF). The specific objectives of this study are as follows: (a) investigate the law of W-RS infiltration, and reveal the reasons for the single-peak curve of IR in W-RS; (b) propose a method of dividing the water infiltration stages in W-RS; (c) develop a unified model that demonstrates the monotonous reduction of IR in HS and reflects the single-peak IR curve for W-RS; (d) explore the differences

and relationship between the proposed models (BF and GMF) and traditional water infiltration models (Philip, Horton, and Kostikov models).

2 | MATERIALS AND METHODS

2.1 | Test material

The test soil samples include HS, which has not been disturbed by human activities, and W-RS with a high humus content. These were taken from the surface layer of a mountainous area in the Guishui River Basin (see Figure 1). The soil particle size (SPS) was measured using a Malvern 2000 laser particle size analyser (Master-sizer Instruments Ltd., UK). The organic matter content (OMC) was measured via oxidation with a potassium dichromate solution ($K_2Cr_2O_7$) in the presence of concentrated sulphuric acid (H_2SO_4). The soil total nitrogen (TN) and total phosphorus (TP) were measured with a UPG-722 ultraviolet spectrophotometer (Beijing Upu Technology Co., Ltd., China). Soil pH was measured with an MIK-PH6.0 m (Hangzhou Asmik Sensing Technology Co., Ltd., China). The soil bulk density (SBD) was measured (Erbach, 1987) using the oven dry method, whereby an intact sample ring (UMS No. 006895, volume 250 mL, inner diameter 80 mm, height 50 mm) is placed into an oven and dried for 24 hr at 105°C. The soil saturated hydraulic conductivity (Ks) was measured with an UMS KSAT device (Metergroup, Inc., Pullman, WA) based on the German standards DIN 19683-9 and DIN 18130-1 and using Darcy's equation; the saturation time of HS was 24 hr and that of W-RS was 48 hr. The soil water repellency was quantified with the water drop penetration time (WDPT) (Dekker, Ritsema, Oostindie, Moore, & Wesseling, 2009). The soil samples were air-dried for 1 week indoors, then droplets of deionized water were placed on flattened subsamples with a pipette (0.48 mL/drop), and the complete infiltration time was measured. The mean time over seven droplet infiltrations was taken as the WDPT value. The measured physicochemical properties (P-CP) of the samples are listed in Table 1.

2.2 | Experimental setup

The soil samples were air-dried and sieved through a 2-mm screen to remove roots, stones, and debris. The HS was then evenly loaded into three cylindrical polymethyl methacrylate (PMMA) boxes (diameter 20 cm, height 100 cm) to an SBD of 1.367 g/cm³, and the W-RS was evenly loaded into three PMMA boxes with the same structure and size to an SBD of 0.874 g/cm³. To reduce the influence of soil air-resistance on the water infiltration process, small holes of diameter 1 cm were opened on one side of the soil column at intervals of 5 cm for soil exhaust. The initial soil water content θ (ISWC) of the HS was set to 4, 8 and 12% in the three samples, respectively, and that of the W-RS was set to 4.7, 6.2, and 9.6%, respectively.

2.3 | Experimental test system

The water infiltration measurement system consists of a Mariotte bottle (used to control the water head constant; radius 5 cm, height 50 cm), cylindrical transparent Perspex chamber (radius 10 cm, height 100 cm), an adjustable lifting table (Wuhan Red Star Yang Science & Technology Co., Ltd., China) to adjust the water head between the soil surface and the water outlet of the Mariotte bottle, a Meacon gravity sensor (Hangzhou Meacon Automation Technology Co., Ltd., China), an RX9600 data collector and Mik-Bsqd signal amplifier (Hangzhou Meacon Automation Technology Co., Ltd., China), and a computer, as shown in Figure 2. In the experiments, the scanning time interval was 2 s and the water head was 1 cm at the soil surface.

2.4 | Mathematical theory

For HS, the Kostikov (Hasan, Chowdhury, Drabo, Kassu, & Glenn, 2015), Philip (Philip, 1992), and Green-Ampt (Selker & Assouline, 2017) models are commonly used to describe infiltration. The Kostikov water infiltration model is derived as follows.

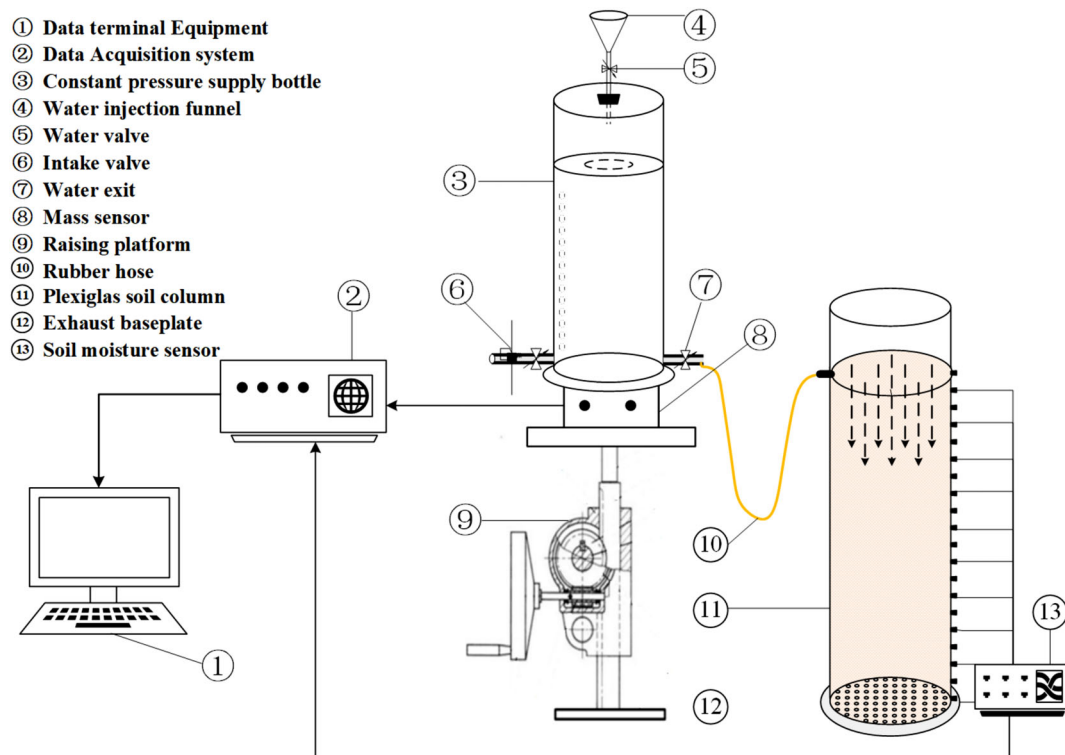


FIGURE 1 Test soils

TABLE 1 Particle size distribution (%) and physicochemical properties of test soils

SPS/mm	0 ~ 0.002	0.002 ~ 0.005	0.005 ~ 0.01	0.01 ~ 0.05	0.05 ~ 0.1	0.1 ~ 0.5	0.5 ~ 2
HS	10.467	7.003	8.946	45.544	20.543	6.853	0.644
W-RS	1.834	1.706	3.263	28.237	35.002	28.324	1.634
P-CP	OMC (g/kg)	TN (g/kg)	TP (g/kg)	pH	SBD (g/m ³)	Ks (cm/min)	WDPT (s)
HS	9.584	0.824	0.063	7.834	1.367	0.0514	0.8031
W-RS	51.982	2.076	0.190	8.473	0.874	0.1292	69.482

Abbreviations: HS, hydrophilic soil; Ks, soil saturated hydraulic conductivity; OMC, organic matter content; P-CP, physicochemical properties; SBD, soil bulk density; SPS, soil particle size; TN, total nitrogen; TP, total phosphorus; WDPT, water drop penetration time; W-RS, water-repellent soil.

**FIGURE 2** Automatic test system of soil water infiltration

2.4.1 | KF/PKF infiltration model

Kostiakov (1932) suggested the following equation for infiltration rate (IR) $i(t)$ as a function of time:

$$i(t) = ant^{n-1}. \quad (1)$$

The piecewise IR function is as follows:

$$i(t) = \begin{cases} a_f n_f t^{n_f-1}, & 0 < t < t_p \\ a_b n_b t^{n_b-1}, & t \geq t_p \end{cases}, \quad (2)$$

where t is time (min); $i(t)$ is the IR at time t (cm/min); a and n are empirical constants that have no physical meaning; a_f and n_f are parameters before the inflection point, a_b and n_b are parameters after the inflection point.

Because the CI curves of W-RS have a double slope (Moret-Fernández et al., 2019) and the IR is a single-peak curve (Ren et al., 2018), the traditional water infiltration model does not provide a good fit (Moret-Fernández et al., 2019). Thus, this study attempts to fit the IR using GF, PGF, FSF, GMF, BF, and PBF. The water infiltration formulas for each function are derived in the following sections.

2.4.2 | GF/PGF infiltration model

Gauss mentioned several concepts related to the Gaussian distribution in his famous book, "Theoria motus corporum coelestium in sectionibus conicis solem ambientium" (Perthes, 2010). The probability density function (PDF) of the GF is shown in Figure 3a, and the form of the GF and PGF (Ren et al., 2018) are as follows:

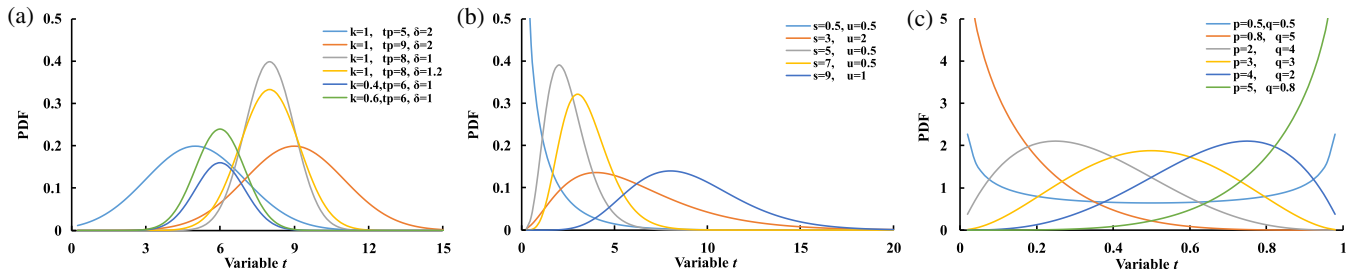


FIGURE 3 PDF of (a) GF, (b) GMF (Equation (7)), and (c) PDF of BF. BF, Beta function; GF, Gauss function; GMF, Gamma function; PDF, probability density function

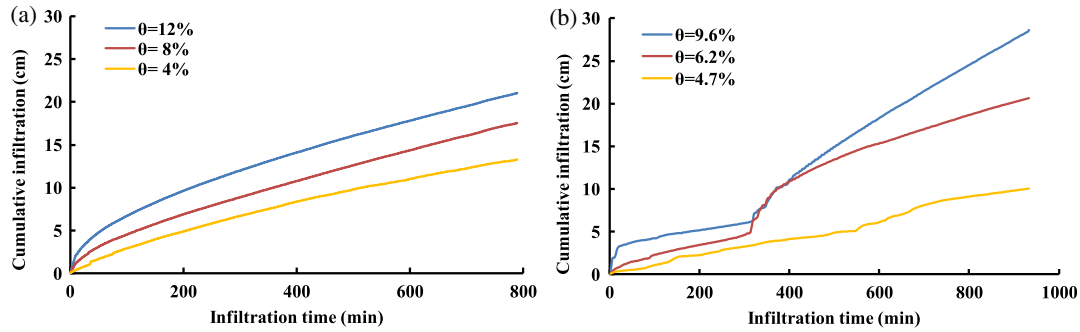


FIGURE 4 Changes in CI over time. (a) HS; (b) W-RS. CI, cumulative infiltration; HS, hydrophilic soil; W-RS, water-repellent soil

$$GF: i(t) = k \cdot \exp\left(-\left(\frac{t-t_p}{\delta}\right)^2\right) + \psi, \quad (3)$$

$$PGF: i(t) = \begin{cases} k_f \cdot \exp\left(-\left(\frac{t-t_p}{\delta_f}\right)^2\right) + \psi_f, & 0 < t < t_p \\ k_b \cdot \exp\left(-\left(\frac{t-t_p}{\delta_b}\right)^2\right) + \psi_b, & t \geq t_p \end{cases} \quad (4)$$

where k and t_p are the coefficient time average and standard deviation of GF, respectively; ψ is a constant; k_f , t_p , δ_f , and ψ_f are the parameters of the PGF before the inflection point; k_b , t_p , δ_b , and ψ_b are the parameters of the PGF after the inflection point.

2.4.3 | FSF infiltration model

Fourier and Freeman (2009) proposed that any function can be expanded into an infinite series of trigonometric functions. In this article, we presume that the IR $i(t)$ has the following form:

$$i(t) = a_0 + \sum_{i=1}^n (a_i \sin(t \cdot \omega) + b_i \cos(t \cdot \omega)), \quad (5)$$

where n is the order number; a_i is the FSF sinusoidal coefficient; b_i is the FSF cosine function coefficient; a_0 is a constant; and ω is the angular frequency.

2.4.4 | GMF infiltration model

GMFs were derived by Euler to solve the problem of extending factorials to the set of real numbers (Davis, 1959). The PDF of a typical GMFs is shown in Figure 3b; the general formula of the PDF is as follows:

$$f(t|s) = \begin{cases} \frac{t^{s-1} e^{-t}}{\Gamma(s)} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (6)$$

Substituting $t = \mu x$ into Equation (6), the PDF of the GMF can be rewritten as:

$$f(t|s, \mu) = \begin{cases} \frac{\mu^s t^{s-1} e^{-\mu t}}{\Gamma(s)} & t \geq 0 \\ 0 & t < 0 \end{cases}, \quad (7)$$

where $f(t|s, \mu)$ is a PDF; s is the shape parameter that controls the amplitude of the curve; and μ is a scale parameter that controls the width of the curve.

The IR of W-RS gradually decreases in the initial stage, and then forms a single-peak curve that first increases and then decreases, that is, the IR is a U-shaped curve from the beginning to the peak (see Figure 5b). To reflect this change, we introduce the

critical time t_0 into Equation (7) to obtain the following modified formula for the IR $i(t)$:

$$i(t) = \eta \frac{\mu^s |t - t_0|^{s-1}}{\Gamma(s)} e^{-\mu|t - t_0|} + \sigma, \quad (8)$$

where t_0 is the time at which the IR reaches the bottom of the valley in the U-shaped curve (min); η is a model coefficient; and σ is a constant.

2.4.5 | BF/PBF infiltration model

The Beta distribution refers to a set of continuous probability distributions defined in the interval (0,1). A random variable x obeys the Beta distribution with parameters p, q (Eugene, Lee, & Famoye, 2002). The BF (Gupta & Nadarajah, 2004) in the interval (0, ∞) is given by:

$$B(p, q) = \int_0^{\infty} \frac{x^{p-1}}{(1+x)^{p+q}} dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}. \quad (9)$$

The PDF of Equation (9) is (Kipping, 2013):

$$f(t|p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} t^{p-1} (1-t)^{q-1}, \quad (10)$$

and its PDF properties are shown in Figure 3c. We assign Equation (10) the coefficient μ and the constant φ , and use it to calculate the IR $i(t)$ of W-RS in the following form:

$$i(t) = \mu \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} t^{p-1} (1-t)^{q-1} + \varphi. \quad (11)$$

Figure 3c shows that the PDF of the BF is in the range [0, 1]. Because water infiltration lasts a long time ($t > 1$), it is necessary to normalize the infiltration time t . We define $t_s = t/t_e$ so that the IR $i(t)$ of the BF and PBF can be written in normalized form as:

$$i(t_s) = \mu \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} t_s^{p-1} (1-t_s)^{q-1} + \varphi, \quad (12)$$

$$i(t_s) = \begin{cases} \mu_f \frac{\Gamma(p_f)\Gamma(q_f)}{\Gamma(p_f+q_f)} t_s^{p_f-1} (1-t_s)^{q_f-1} + \varphi_f, & 0 < t < t_p \\ \mu_b \frac{\Gamma(p_b)\Gamma(q_b)}{\Gamma(p_b+q_b)} t_s^{p_b-1} (1-t_s)^{q_b-1} + \varphi_b, & t \geq t_p, \end{cases} \quad (13)$$

where t_e is the end of the infiltration time (min); t_s is the normalization time; p and q are the shape and scale parameters; μ is the model coefficient; and φ is a constant. μ_f, p_f, q_f , and φ_f are the model parameters before the inflection point, μ_b, p_b, q_b , and φ_b are the parameters after the inflection point.

Equation (1) is a monotonic decreasing curve when $n - 1 < 0$, that is, $n < 1$, which can be used to fit the IR of HS. In Equation (3), k affects

the height of the curve, t_b is the temporal inflection point, which affects the distribution of the crest on the timeline, and δ affects the width of the crest. The GF of Equation (3) can be used to fit the IR of W-RS when the stable IR after the inflection point is equal to that before the inflection point. In Equation (5), n is the order number affecting the number of crests; a_i and b_i jointly determine the height of the curve; and ω is the angular frequency affecting the width of the crest. The FSF can be used to fit the IR of W-RS when the IR has multiple crests. BF is a monotonic decreasing curve when $0 < s \leq 1$, and can be used to fit the IR of HS. Its PDF is a single-peak curve when $s > 1$, and can be used to fit the IR of W-RS. The PDF of BF is a U-shaped curve when $0 < p < 1$ and $0 < q < 1$; when $0 < p \leq 1$ and $q > 1$, the PDF is a monotonic decreasing curve that can be used to fit the IR of HS; when $p > 1$ and $0 < q \leq 1$, the PDF is a monotonic increasing curve, and when $p > q > 1$, the PDF is a left-skew distribution curve; when $q > p > 1$, the PDF is a right-skew distribution curve that can be used to fit the IR of W-RS.

3 | RESULTS

3.1 | Changes in CI and IR over time

The changes in CI over time for the HS and W-RS samples are shown in Figure 4, and the change in IR over time is shown in Figure 5. Because of the influence of the ordinate axis scale, the CI and IR changes do not appear to be significant when $\theta = 4.7\%$ (Figures 4b and 5b). Thus, we scaled the ordinate of CI and IR, and present the enlarged plots in Figure 6.

Figure 4 shows that higher values of ISWC produce a larger CI in both the HS and W-RS samples. The difference is that CI exhibits smooth monotonic growth for HS (Figure 4a), but a more variable increase for W-RS. The WR effect is very evident in these CI curves, similar to those measured by Moret-Fernández et al. (2019), where the CI has an inflection point and demonstrates smooth monotonic growth at the critical point (Figure 4b). Lichner, Capuliak, et al. (2013) and Lichner, Hallett, et al. (2013) defined the relationship between CI and the square root of time as "hockey-stick-like", and suggested a procedure for estimating the WR cessation time. Figure 5 shows that larger values of ISWC give a larger IR. The IR of the HS sample decreases monotonously with time (Figure 5a) because of the decreasing contribution of the capillary (or pressure head)-driven component of the hydraulic gradient that develops as the wetting front moves further away from the surface (Lichner, Capuliak, et al. 2013; Lichner, Hallett, et al. 2013). The IR then approaches steady infiltration. In contrast, the IR of the W-RS sample (Figure 5b) exhibits a gradual decrease in the initial period (0–20 min), and then presents a single-peak curve that increases before decreasing. The larger ISWC values ($\theta = 6.2\%$ and $\theta = 9.6\%$) produce an earlier inflection point ($\cong 310$ min) than the smaller ISWC ($\theta = 4.7\%$, $\cong 568$ min). For all three ISWCs, the wave peak of the IR is narrow and the process of reaching the maximum IR is dramatic, and the attenuation process is slower after the peak. That is, the IR presents an asymmetric single-peak

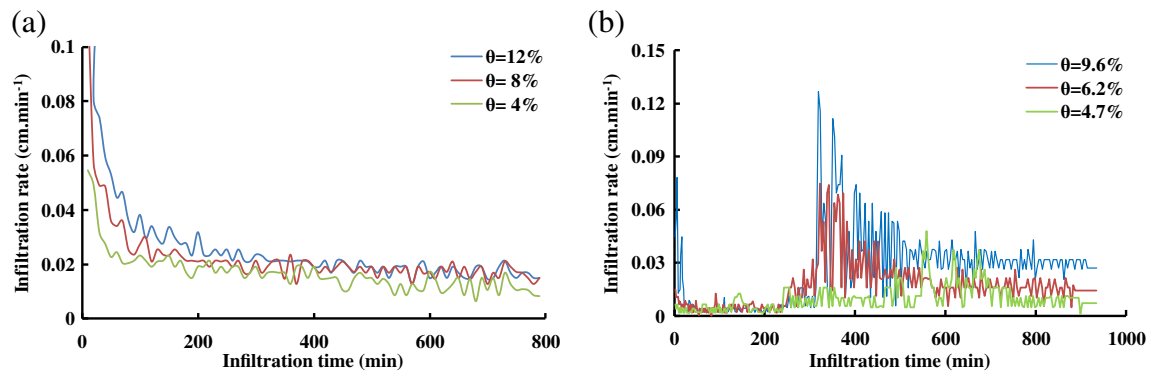


FIGURE 5 Changes in IR over time. (a) HS; (b) W-RS. HS, hydrophilic soil; IR, infiltration rate; W-RS, water-repellent soil

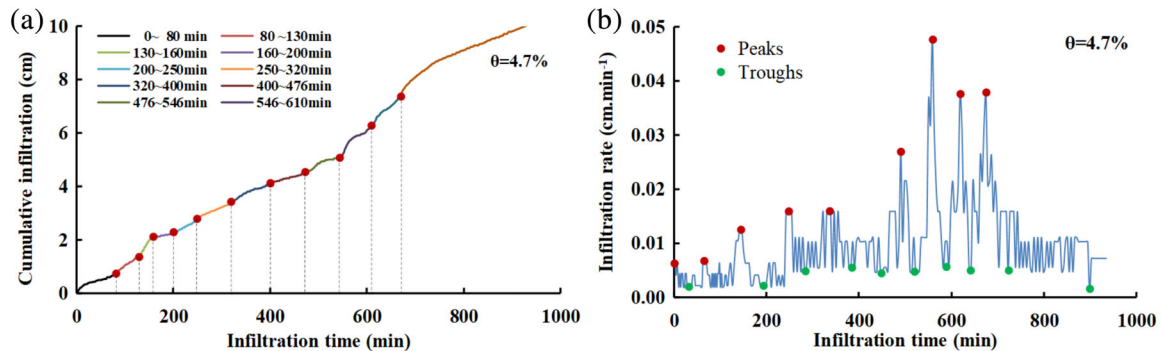


FIGURE 6 Changes in (a) CI and (b) IR over time for W-RS when $\theta = 4.7\%$. CI, cumulative infiltration; IR, infiltration rate; W-RS, water-repellent soil

curve with a left-skewed distribution. From Figure 5b, we can observe that the stable IR after the peak is larger than that before the peak. In other words, the water enters the soil slowly before the inflection point, and then enters more quickly after this point. This result is in accordance with that reported by Moret-Fernández et al. (2019). The slower infiltration before the inflection point could be explained by the hydrophobicity preventing water from entering the soil, resulting in a low IR. In essence, the hydraulic gradient is mainly determined by the matrix in unsaturated flow (Vogelmann et al., 2017), and WR reduces the matric potential. Hence, the volume of water-conductive pores is lower, promoting a decrease in IR. Following the wetting of hydrophobic compounds, the repellency effect disappears after prolonged contact with water (Filipović et al., 2018), allowing the IR to increase as the WR disappears (Alagna, Iovino, Bagarello, Mataix-Solera, & Lichner, 2017). Another viewpoint is that any initially postponed infiltration increases after the CA between the water and the soil particles decreases (Bughici & Wallach, 2016). The mechanisms for IR increase suggested by Alagna et al. (2017) and Bughici and Wallach (2016) are basically the same when stated in terms of CA.

The CI curve is relatively smooth before and after the inflection point (Figure 4b), and the IR has a significant peak when $\theta = 6.2\%$ and $\theta = 9.6\%$ (Figure 5b). However, Figure 6a shows that the CI has several inflection points and a distinctly mutated inflection point at 580 min when $\theta = 4.7\%$. In this case, the CI exhibits a multi-stage growth trend. This is consistent with the results of Filipović et al. (2018) using

HYDRUS (2D/3D) to simulate the hydraulic performance of W-RS under drought conditions.

Figure 6b shows that there are multiple peaks and troughs on either side of the maximum IR (580 min). This suggests that the infiltration process of W-RS has unstable infiltration properties when the ISWC value is small, which may be caused by spatial variability in the WR (Liu, Rahav, & Wallach, 2019) leading to preferential flow characteristics in WR soils (Rye & Smettem, 2017). Because of the irregular distribution of the priority path, the infiltration properties are unstable. Unstable infiltration may also be caused by the surface pressure head being less than the water-entry value, which is positive in repellent soils (Wang et al., 2000). However, taking 580 min as the cutoff when $\theta = 4.7\%$, the IR gradually increases before and gradually decreases after this point. From a macro perspective, this is still a single-peak curve.

Wang et al. (1998) suggested that when the air pressure ahead of the wetting front reaches an air-breaking value, soil air escapes from the surface, leading to an immediate decrease in the air pressure and an increase in the infiltration rate. When the air pressure falls below a certain air-closing value, air escape stops, the infiltration rate decreases, and the air pressure increases. This cyclic process repeats itself during the entire infiltration period, possibly explaining the inflection point in CI (Figures 5b and 6a) and significant variability in IR (Figures 5b and 6b), which are inconsistent with the traditional infiltration theory.

For the W-RS, before the experiment, we adjusted the water head difference (water-ponding depth) h_0 between the Markova bottle and soil surface to 1 cm. After a period of infiltration, the soil surface was 2 cm higher than the initial surface as a result of volume expansion (Figure 14a). The soil water was considered to be within 1 cm positive pressure infiltration for a short time after the beginning of the experiment, followed by $h_0 = 1 \text{ cm} - 2 \text{ cm} = -1 \text{ cm}$ negative pressure with a water-entry value of $h_{we} > 0$ (Wang et al., 2000). Therefore, the water head difference Δh between the water-ponding depth and the water-entry pressure head satisfied some pressure criterion, and when $\Delta h = h_0 - h_{we} = -1 \text{ cm} - h_{we} < 0$, unstable flow occurred (Wang et al., 1998).

3.2 | Relevance analysis

Equations (1), (3), (5), (8), and (12) were used to calculate the IR of the HS sample. The measured and calculated values are shown in Figure 8; the fitting formulas, determination coefficient (R^2), and root mean square error (RMSE) between the measured and calculated values are presented in Table 2. Equations (2)–(5), (8), and (13) were used to calculate the IR of the W-RS sample. For the five models, the R^2 and RMSE between the measured and calculated IRs are listed in Tables 3 and 4, and the measured and calculated values of IR are shown in Figures 9–13, which shows the maximal IR (peak between stages III and IV) increases with an increase in the initial soil water content. The MATLAB Curve Fitting Toolbox was used to solve the

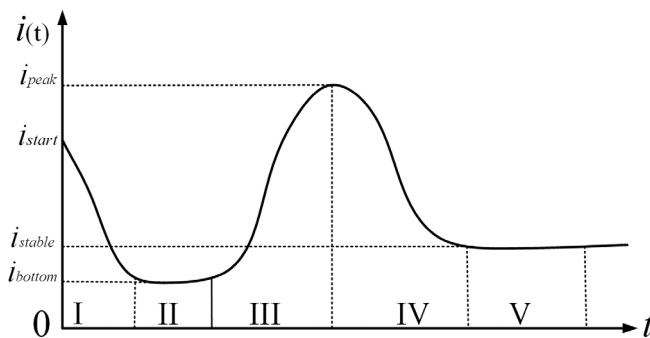


FIGURE 7 Infiltration procedure of water repellent soil

KF/PKF, GF/PGF, and FSF water infiltration models with the Levenberg–Marquardt algorithm, Robustness set to LAR, MaxIter = 400, and default values of the other parameters. The Optimization Toolbox was used to solve the BF/PBF and GMF water infiltration models, with a genetic algorithm selected for the parameter inversion method, Popsiz = 50, Generations (MaxIter) = 100, Migration Fraction = 0.85, Crossover Fraction = 0.15, Elite count = 5, and default values for the other parameters.

For HS, the BF model gives the largest average R^2 value, followed by GMF, KF, GF, and FSF in that order; the maximum R^2 value is 0.9464 (BF) and the minimum value is 0.7141 (FSF). The lowest mean RMSE is given by FSF, followed by KF, GF, GMF, and BF in that order; the minimum RMSE value is 0.01178 (FSF), and the maximum value is 0.01636 (BF). For the HS sample, GMF, KF, and GF have relatively good fitting accuracy, whereas GF and FSF provide relatively poor fits. Tables 3 and 4 show that the GMF model achieved the highest average R^2 between the measured and calculated IR for the W-RS sample, followed by FSF, PKF, BF, PBF, PGF, and GF in that order. The maximum R^2 is 0.7082 (GMF) and the minimum is 0.5429 (GF). The smallest mean RMSE is given by the GMF model, followed by PKF, FSF, PBF, PGF, and GF in that order. The minimum RMSE is 0.00543 (GMF) and the maximum is 0.01852 (GF). For the W-RS sample, GMF and PKF provide a relatively good fit, FSF and PBF give a reasonable fit, whereas PGF and GF PF are poorly fitted.

The inflection point, which indicates the WR cessation time (Moret-Fernández et al., 2019), was estimated from the point of intersection of two straight lines, representing the CI relationships for the W-RS and the square root of time (Lichner, Capuliak, et al. 2013; Lichner, Hallett, et al. 2013). From Table 3 (rows 2–4, column 3), we can see that the GF parameter t_p (581, 352, and 326) is close to the CI inflection point (Figure 4b), which corresponds to the maximum IR (Figure 5b). Compared with the method of using the intersection of two straight lines to estimate the WR cessation time, the GF provides a direct and concise approach.

3.3 | W-RS infiltration process

Unlike the HS infiltration process, which is divided into rapid infiltration, gradual decrease, and steady infiltration stages, the shape of

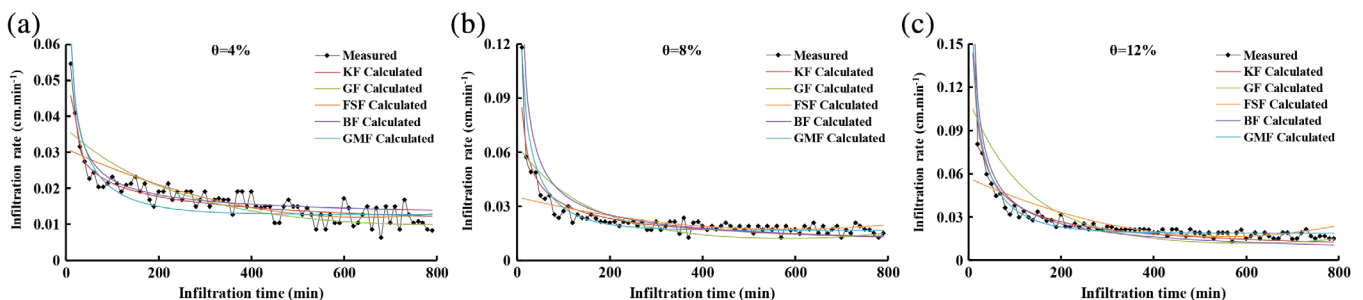


FIGURE 8 Measured and calculated values of IR for HS sample. (a) $\theta = 4\%$; (b) $\theta = 8\%$; (c) $\theta = 12\%$. HS, hydrophilic soil; IR, infiltration rate

TABLE 2 Parameter fitting of IR using KF, GF, FSF, GMF, and BF

Infiltration model	IWC (%)	Fitting formula of IR	R ²	RMSE
KF	4	$i(t) = 0.0804t^{-0.2805}$	0.9281	0.00639
	8	$i(t) = 0.2121t^{-0.4149}$	0.9246	0.01253
	12	$i(t) = 0.5920t^{-0.5843}$	0.9408	0.02236
GF	4	$i(t) = 0.0695 \cdot \exp\left(-\left(\frac{t+864}{1027}\right)^2\right) + 0.0065$	0.8251	0.00700
	8	$i(t) = 0.3069 \cdot \exp\left(-\left(\frac{t+817}{1510}\right)^2\right) + 0.0087$	0.7694	0.01273
	12	$i(t) = 0.0444 \cdot \exp\left(-\left(\frac{t+737}{1138}\right)^2\right) + 0.0091$	0.7672	0.02345
FSF	4	$i(t) = 283900.031 - 104.2\sin(5.708 \times 10^{-7} \cdot t) + 283900\cos(5.708 \times 10^{-7} \cdot t)$	0.8002	0.00631
	8	$i(t) = 172100.035 - 78.57\sin(7.888 \times 10^{-7} \cdot t) + 172100\cos(7.888 \times 10^{-7} \cdot t)$	0.6783	0.01005
	12	$i(t) = 313600.570 - 159.2\sin(9.103 \times 10^{-7} \cdot t) + 313600\cos(9.103 \times 10^{-7} \cdot t)$	0.6635	0.01898
GMF	4	$i(t) = 7 \frac{100^{0.5}t^{-0.5}}{\Gamma(0.5)} e^{-100t} + 0.00165$	0.8912	0.00728
	8	$i(t) = 8 \frac{107^{0.5}t^{-0.5}}{\Gamma(0.5)} e^{-107t} + 0.00195$	0.9620	0.01418
	12	$i(t) = 9 \frac{126^{0.5}t^{-0.5}}{\Gamma(0.5)} e^{-126t} + 0.00219$	0.9526	0.02234
BF	4	$i(t) = 0.1842 \frac{\Gamma(0.35)\Gamma(1.2)}{\Gamma(1.55)} \left(\frac{t}{800}\right)^{-0.65} \left(1 - \frac{t}{800}\right)^{0.2} + 0.0107$	0.9176	0.00712
	8	$i(t) = 0.3227 \frac{\Gamma(0.45)\Gamma(1.2)}{\Gamma(1.65)} \left(\frac{t}{800}\right)^{-0.55} \left(1 - \frac{t}{800}\right)^{0.2} + 0.0053$	0.9637	0.01711
	12	$i(t) = 0.4796 \frac{\Gamma(0.55)\Gamma(1.2)}{\Gamma(1.75)} \left(\frac{t}{800}\right)^{-0.45} \left(1 - \frac{t}{800}\right)^{0.2} + 0.0012$	0.9580	0.02486

Abbreviations: BF, Beta function; FSF, Fourier series function; GF, Gauss function; GMF, Gamma function; IR, infiltration rate; KF, Kostiakov function; RMSE, root mean square error.

Figure 5b suggests that the W-RS infiltration process should be divided into five sections (Figure 7). In *Phase I*, the IR decreases (*stage I* of Figure 7). Because the soil air pressure is almost zero at the initial point, there is a local maximum IR at i_{start} . In *Phase II*, the infiltration process is temporarily stable before the inflection point (*stage II* of Figure 7), because the air-resistance of the deep soil layer, the repulsive force of the water repellency, and the water gravitational and matric potential are in a state of equilibrium. There is a minimum IR of i_{bottom} , and the IR caused by the compact layer gradually decreases. In *Phase III*, the IR increases (*stage III* of Figure 7), possibly because of a gradual decrease in WR. When the soil water content exceeds the critical value for the disappearance of WR, there is a maximum IR of i_{peak} . In *Phase IV*, the IR decreases (*stage IV* of Figure 7), possibly because the infiltration is governed by gravity and the pressure head gradient, which diminishes as the wetting front deepens. Grismer, Orang, Clausnitzer, and Kinney (1994) suggested that the water pressure gradient across the transmission zone decreases because the soil air pressure balances the capillary suction at the wetting front. Decreases in both the hydraulic conductivity and the pressure gradient combine to reduce the infiltration rate. In *Phase V*, the infiltration process stabilizes after the inflection point (*stage V* of Figure 7), and the IR reaches a value of i_{stable} . This may be caused by the gravitational potential of free water in the upper saturated soil layer being greater than the soil air-resistance, similar to the infiltration flow becoming stable if the ponding head is greater than the soil water-entry value (Wang et al., 2000). Ignoring the initial section of rapid water infiltration, the W-RS IR is a right-skewed single-peak curve.

3.4 | Graphical analysis

Figure 8 shows that, when using FSF and GF for HS, the calculated IR values are larger than the measured values in the early stage, but coincide well in the later stage. However, KF, GMF, and BF exhibit good fitting effects in both the early and late stages.

Figure 9 shows that the IR simultaneously has a maximum and minimum at the inflection point when the W-RS is fitted by PKF, for which the IR mutation is almost instantaneous. In fact, because WR is affected by the ISWC, the disappearance of WR is a slow process (Filipović et al., 2018). As the soil water content increases, once the WR vanishes, all soil pores start to participate in the infiltration process, producing an increase in IR (Moret-Fernández et al., 2019). PKF not only describes the gradual decrease in IR from i_{start} to i_{bottom} (*stage I* of Figure 7) and from i_{peak} to i_{stable} (*stage IV* of Figure 7), but also reflects the phenomenon whereby the steady IR after the inflection point (*stage V* of Figure 7) is greater than that before the inflection point (*stage II* of Figure 7). However, it is difficult to reflect the gradual increase in IR from i_{bottom} to i_{peak} (*stage III* of Figure 7).

Figure 10 shows that the IR of W-RS increases and then decreases with a single peak when calculated by GF. However, because IR is a bell-shaped symmetric figure with an inflection point when fitted by GF, it is difficult to reflect the phenomenon in which the stable IR after the inflection point (*stage V* of Figure 7) is greater than that before the inflection point (*stage II* of Figure 7), and cannot reproduce the gradual decrease in IR in the initial section (*stage I* of Figure 7). PGF also struggles to reflect the gradual decrease in IR in the initial stages (*stage I* of Figure 7), but can mimic the single-peak

TABLE 3 Parameter fitting of IR by GF, GMF, and fourth-order FSF

Infiltration model	IWC/%	Fitting formula of IR	R ²	RMSE
GF	4.7	$i(t) = 0.0475 \cdot \exp\left(-\left(\frac{t-581}{127}\right)^2\right) + 0.0047$	0.5523	0.01140
	6.2	$i(t) = 0.0715 \cdot \exp\left(-\left(\frac{t-352}{117}\right)^2\right) + 0.0963$	0.6102	0.01758
	9.6	$i(t) = 0.1408 \cdot \exp\left(-\left(\frac{t-326}{92}\right)^2\right) + 0.0236$	0.4663	0.02651
GMF	4.7	$i(t) = 12.5 \frac{34^{12} t-180 ^{11}}{\Gamma(12)} e^{-34 t-180 } + 0.00395$	0.6782	0.00635
	6.2	$i(t) = 17.2 \frac{40^{6.7} t-113 ^{5.7}}{\Gamma(6.7)} e^{-40 t-113 } + 0.00941$	0.7131	0.00542
	9.6	$i(t) = 27.1 \frac{38^7 t-125 ^6}{\Gamma(7)} e^{-38 t-125 } + 0.01885$	0.7334	0.00453
Fourth-order FSF	4.7	$i(t) = 0.0083 + 0.0004 \cdot \sin(0.0059 \cdot t) - 0.0049 \cdot \sin(0.0059 \cdot t) - 0.0003 \cdot \sin(0.0059 \cdot t) - 0.0011 \cdot \sin(0.0059 \cdot t) - 0.0025 \cdot \cos(0.0059 \cdot t) + 0.0021 \cdot \cos(0.0059 \cdot t) - 0.0031 \cdot \cos(0.0059 \cdot t) + 0.0009 \cdot \cos(0.0059 \cdot t)$	0.6246	0.00974
	6.2	$i(t) = 0.0212 + 0.0121 \cdot \sin(0.0084 \cdot t) - 0.0155 \cdot \sin(0.0084 \cdot t) - 0.0064 \cdot \sin(0.0084 \cdot t) + 0.0035 \cdot \sin(0.0084 \cdot t) - 0.0037 \cdot \cos(0.0084 \cdot t) + 0.0063 \cdot \cos(0.0084 \cdot t) - 0.0049 \cdot \cos(0.0084 \cdot t) - 0.0029 \cdot \cos(0.0084 \cdot t)$	0.7612	0.01475
	9.6	$i(t) = 0.0264 + 0.0040 \cdot \sin(0.0065 \cdot t) + 0.0056 \cdot \sin(0.0065 \cdot t) - 0.0128 \cdot \sin(0.0065 \cdot t) - 0.0036 \cdot \sin(0.0065 \cdot t) - 0.0077 \cdot \cos(0.0065 \cdot t) + 0.0013 \cdot \cos(0.0065 \cdot t) - 0.0128 \cdot \cos(0.0065 \cdot t) - 0.0011 \cdot \cos(0.0065 \cdot t)$	0.6772	0.02219

Abbreviations: FSF, Fourier series function; GF, Gauss function; GMF, Gamma function; IR, infiltration rate; RMSE, root mean square error.

process of IR increasing and then decreasing (stages III–IV of Figure 7), and can also reflect the phenomenon whereby the stable IR after the inflection point (stage V of Figure 7) is greater than that before the inflection point (stage II of Figure 7).

For the W-RS sample, Figure 11 shows that the IR has a single-peak curve when calculated by a fourth-order FSF. However, there are multiple peaks on both sides of the inflection point, which is inconsistent with the theory that the IR gradually increases before the inflection point and gradually decreases after the inflection point. Therefore, FSF does not accurately reflect the infiltration procedure of W-RS.

Figure 12 shows the results of using the GMF to calculate the IR of the W-RS sample. The IR differs slightly before and after the stable peak, and the measured IR is in good agreement with the calculations when $\theta=4.7\%$. Although the GMF struggles to reflect the stable IR after the inflection point (stage V of Figure 7) being greater than that before the inflection point (stage II of Figure 7) when $\theta=6.2\%$ and $\theta=9.6\%$, it produces the gradual decrease in IR after the beginning of infiltration (stage I of Figure 7) as well as the single-peak process of increasing and then decreasing IR (stages III–IV of Figure 7) when the WR disappears.

From Figure 13, we can see that the PBF model not only reflects the change in the single-peak curve (stages III–IV of Figure 7), but also produces the phenomenon whereby the stable IR after the inflection point (stage V of Figure 7) is greater than before the inflection point (stage II of Figure 7). The PBF model also gives a gradually decreasing IR in the initial phase of water infiltration (stage I of Figure 7).

In conclusion, for W-RS, PBF not only reflects the monotonous decrease of IR ($0 < p \leq 1, q > 1$) and the U-shaped change in IR before and after the inflection point, but also recreates the right-skewed distribution curve of IR before and after the inflection point, indicating good applicability.

4 | DISCUSSION

4.1 | Reasons for single-peak curve

Figure 5b shows that CI appears to have an inflection point, and Figure 3b shows that IR is a single-peak curve that has a higher value after the peak than before. We think this may be caused by the following factors: (a) *Disappearance of WR*: The water drop penetration time and ISWC have a one-peak distribution (Gao, Lin, Liu, & Wu, 2018), suggesting that WR and ISWC follow a normal distribution. The wetting body ISWC above the wetting front increases gradually with the continuous infiltration of water. When ISWC becomes greater than the critical ISWC (corresponding to the maximum WR), the WR begins to decrease, and the repellency effect disappears after prolonged contact with water (Filipović et al., 2018). Thus, the IR increases over a longer duration after the WR has disappeared (Alagna et al., 2017). (b) *Soil volume expansion*: We conducted rainfall runoff tests and one-dimensional vertical water infiltration experiments using W-RS. The results show that water-repellent humus soil has a degree of expansibility (Figure 14a), with volume expansion of $\sim 17.5\%$ under fully saturated conditions. Higher values of OMC (Table 1, row

TABLE 4 Parameter fitting of IR by PKF, PGF, and PBF

Infiltration model	IWC/%	Fitting formula of IR		R ²	RMSE
		0 < t < t _p	t ≥ t _p		
PKF	4.7	$i(t) = 0.04795t^{-0.2278}$	$i(t) = 0.00059t^{-0.9872}$	0.4582	0.00740
	6.2	$i(t) = 0.00122t^{-0.8970}$	$i(t) = 0.00103t^{-0.9867}$	0.7789	0.01419
	9.6	$i(t) = 0.00588t^{-0.9109}$	$i(t) = 0.00196t^{-0.9855}$	0.7424	0.02369
PGF	4.7	$i(t) = 0.0475 \cdot \exp\left(-\left(\frac{t+581}{127}\right)^2\right) + 0.0052$	$i(t) = 0.0475 \cdot \exp\left(-\left(\frac{t+581}{127}\right)^2\right) + 0.0052$	0.5361	0.01079
	6.2	$i(t) = 0.0736 \cdot \exp\left(-\left(\frac{t+332}{47}\right)^2\right) + 0.0048$	$i(t) = 0.0635 \exp\left(-\left(\frac{t+332}{117}\right)^2\right) + 0.0149$	0.7644	0.01697
	9.6	$i(t) = 0.1317 \cdot \exp\left(-\left(\frac{t+326}{60}\right)^2\right) + 0.0057$	$i(t) = 0.124 \cdot \exp\left(-\left(\frac{t+326}{90}\right)^2\right) + 0.0287$	0.5171	0.02780
PBF	4.7	$i(t) = 0.0062 \frac{\Gamma(27)\Gamma(18)}{\Gamma(45)} \left(\frac{t}{934}\right)^{26} \left(1 - \frac{t}{934}\right)^{17} + 0.0047$	$i(t) = 0.0062 \frac{\Gamma(27)\Gamma(18)}{\Gamma(45)} \left(\frac{t}{934}\right)^{26} \left(1 - \frac{t}{934}\right)^{17} + 0.0047$	0.6094	0.01034
	6.2	$i(t) = 0.0096 \frac{\Gamma(0.4)\Gamma(0.15)}{\Gamma(0.55)} \left(\frac{t}{934}\right)^{-0.6} \left(1 - \frac{t}{934}\right)^{-0.85} + 0.0017$	$i(t) = 0.0126 \frac{\Gamma(0.45)\Gamma(1.5)}{\Gamma(1.95)} \left(\frac{t}{934}\right)^{-0.55} \left(1 - \frac{t}{934}\right)^{0.5} + 0.0114$	0.7462	0.01531
	9.6	$i(t) = 0.0241 \frac{\Gamma(0.5)\Gamma(0.35)}{\Gamma(0.85)} \left(\frac{t}{934}\right)^{-0.5} \left(1 - \frac{t}{934}\right)^{-0.65} + 0.0021$	$i(t) = 0.0139 \frac{\Gamma(0.5)\Gamma(4)}{\Gamma(4.5)} \left(\frac{t}{934}\right)^{-0.5} \left(1 - \frac{t}{934}\right)^3 + 0.0284$	0.6163	0.02652

Abbreviations: IR, infiltration rate; PBF, piecewise Beta function; PGF, piecewise Gauss function; PKF, piecewise Kostiakov function; RMSE, root mean square error.

6, column 2) enhance aggregate stability (Algayer et al., 2014) and increase soil porosity. Volume expansion leads to an increase in porosity, and especially an increase in macropores (Menon et al., 2011), which allow the penetration of water in the saturated surface soil. (c) *Root growth of plants*: Strong root activity in humus soil (Agnelli et al., 2016) means there may be several plant shoots (0–0.5 cm high) on the soil surface, and a large number of white root systems (0–10 cm) growing inside the soil (Figure 14b). When water infiltrates the soil, the biological activity of plant seeds in the soil is activated (Mašková & Weiser, 2019). Thus, the growth of root cells accelerates the absorption of water by soil particles. More importantly, some closed pores in the soil are connected by roots, and infiltration accelerates with preferential flow along those roots. (d) *Increasing soil temperature*: In our experiments, we found that the temperature of wetted soil was significantly higher than that of the dry parts of soil (Figure 14c), and also higher than the water temperature in the Mariotte bottle. The maximum difference in temperature between the wetted and dry regions was 3.4°C (maximum temperature of 18.3°C, minimum temperature of 14.9°C). This indicates that the temperature increases come from biochemical reactions inside the soil. The viscous coefficient of water decreases as the soil temperature increases, thus enhancing the ability of water movement (Ben Amar, Saidani, Deratani, & Palmeri, 2007) and accelerating the infiltration.

In summary, the mutation phenomenon of CI and IR results from the physical structure combined with chemical and biological factors. The results further prove that WR is a transient and spatially distributed physical soil property affecting soil hydrological, chemical, and biological processes (Liu et al., 2019).

4.2 | Model applicability

We now analyse the conditions under which each of the models should be employed. For W-RS, if the initial infiltration process (*stage I* of Figure 14) has a short duration, and the stable IR before the inflection point (*stage II* of Figure 7) is equal to that after the inflection point (*stage IV* of Figure 7), the IR is a symmetrical bell-shaped curve (ignoring the initial short-duration infiltration), so the infiltration process can be modelled by GF. When the short duration of the initial infiltration process is considered, and the stable IR before the inflection point is equal to after the inflection point, GMF produces better simulation results. When the BPF is used to fit the IR of W-RS, the process can be divided into three patterns. *Model 1*: a monotonous decrease in infiltration in the first part (*stages I and II* of Figure 7) with $0 < p \leq 1$ and $q > 1$, and the right-skewed distribution curve infiltration process in the second part (*stages III, IV, and V*), with $1 < q < p$. *Model 2*: U-shaped infiltration in the initial process (*stages I, II, and III*) with $0 < p < 1$, $0 < q < 1$. The initial IR is greater than that at the inflection point when $0 < p < q < 1$, and less than that at the inflection point when $0 < q < p < 1$. IR decreases monotonously in the second part (*stages IV and V*) with $0 < p \leq 1$ and $q > 1$. *Model 3*: IR with a left-skewed distribution curve when $p > q > 1$, and a right-

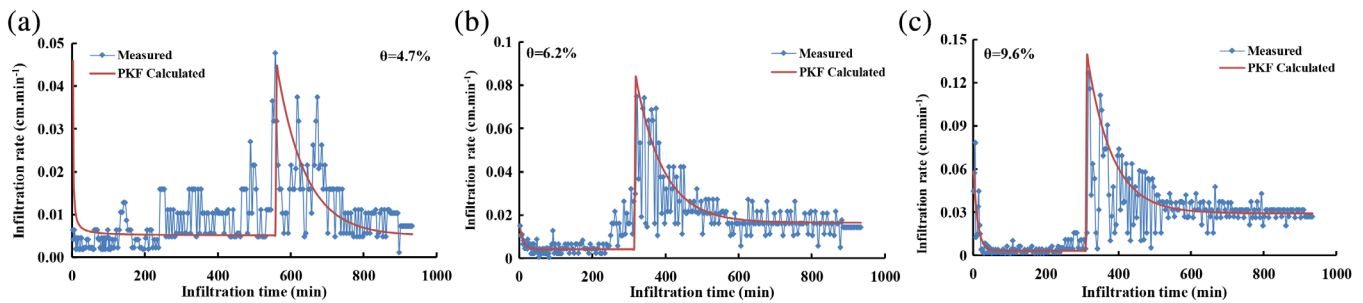


FIGURE 9 Measured and calculated values of IR for W-RS sample using PKF. (a) $\theta = 4.7\%$; (b) $\theta = 6.2\%$; (c) $\theta = 9.6\%$. IR, infiltration rate; PKF, piecewise Kostiakov function; W-RS, water-repellent soil

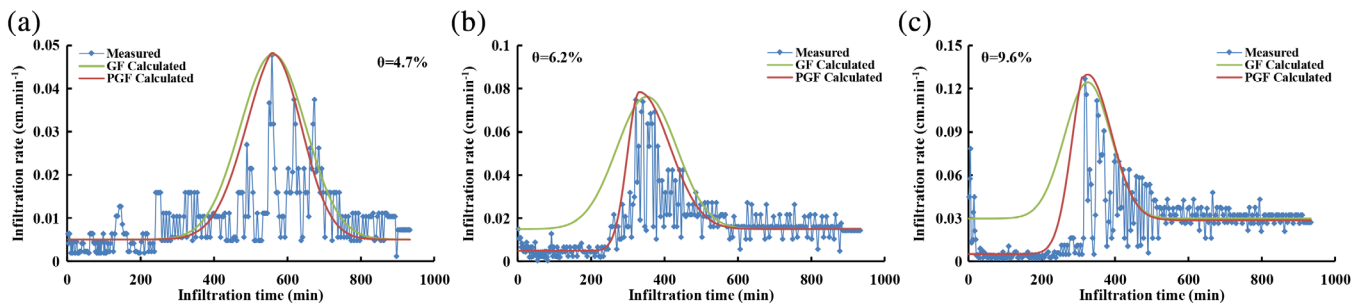


FIGURE 10 Measured and calculated values of IR for W-RS sample using GF and PGF. (a) $\theta = 4.7\%$; (b) $\theta = 6.2\%$; (c) $\theta = 9.6\%$. GF, Gauss function; IR, infiltration rate; PGF, piecewise Gauss function; W-RS, water-repellent soil

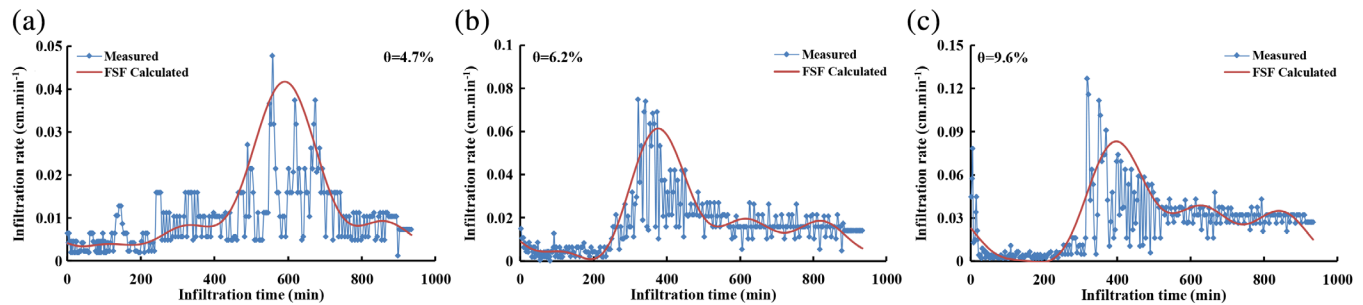


FIGURE 11 Measured and calculated values of IR for W-RS sample using fourth-order FSF. (a) $\theta = 4.7\%$; (b) $\theta = 6.2\%$; (c) $\theta = 9.6\%$. FSF, Fourier series function; IR, infiltration rate; W-RS, water-repellent soil

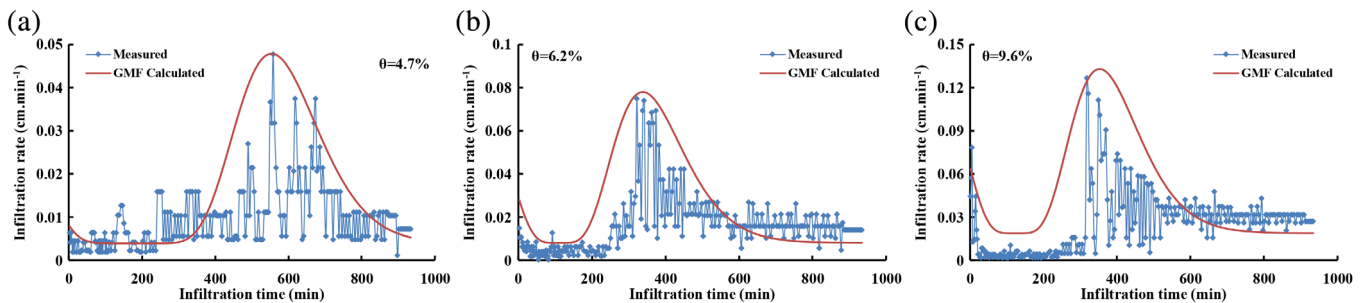


FIGURE 12 Measured and calculated values of IR for W-RS sample using GMF. (a) $\theta = 4.7\%$; (b) $\theta = 6.2\%$; (c) $\theta = 9.6\%$. GMF, Gamma function; IR, infiltration rate; W-RS, water-repellent soil

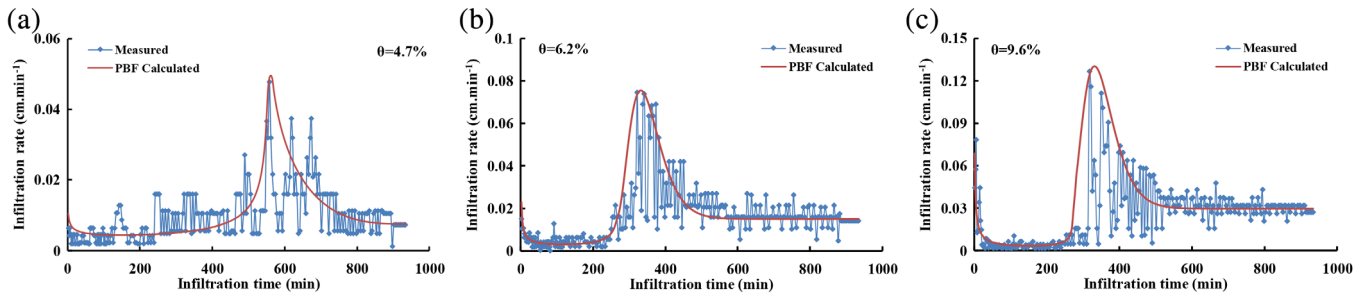


FIGURE 13 Measured and calculated values of IR for W-RS sample using PBF. (a) $\theta = 4.7\%$; (b) $\theta = 6.2\%$; (c) $\theta = 9.6\%$. IR, infiltration rate; PBF, piecewise Beta function; W-RS, water-repellent soil

FIGURE 14 Expansion with watering, development of plant roots and temperature rises in W-RS. (a) Volumetric expansion of W-RS. (b) Development and growth of plant root in W-RS. (c) Temperature rises as a result of physiological and biochemical reactions in W-RS. W-RS, water-repellent soil



skewed distribution curve when $1 < p < q$ under the condition that the IR in stage I can be neglected. In this study, the IR is calculated by model 3 with the BF and a left-skewed distribution when $\theta = 4.7\%$, and by model 2 with the PBF when $\theta = 6.2\%$ and $\theta = 9.6\%$.

The relationship between the mathematical model proposed in this article and the traditional water infiltration model is as follows:

For the GMF water infiltration model given by Equation (8), we define $t_0 = 0$, $s = 1$. As $\Gamma(1) = 1$, $|t - t_0|^{1-1} = 1$, $\mu^1 = \mu$, Equation (8) degenerates into the following mathematical model:

$$i(t) = \eta \mu e^{-\mu t} + \sigma. \quad (14)$$

This is similar to Horton's infiltration model. The coefficients $\eta \mu$ in Equation (14) are equivalent to $i_c - i_0$, and σ is equivalent to i_0 (Beven, 2004; Verma, 1982). Note that i_c is the maximum IR in the initial stages of Horton's infiltration model, whereas μ influences the decrease in the infiltration curve and η influences the overall size of the IR in Equation (14). We consider Equation (14) to be a variation of Horton's model. When $t > 0$, IR is a monotonically decreasing function.

When $t_0 = 0$ and $\mu = 1$, Equation (8) degenerates into a complex exponential function:

$$i(t) = \eta \frac{t^{s-1}}{\Gamma(s)} e^{-t} + \sigma. \quad (15)$$

Numerical simulations show that when $s > 4.907$, Equation (15) is a monotonically decreasing function; for $0 < s < 4.907$, Equation (15) is a single-peak function.

For the BF water infiltration model in Equation (12), we define $P = 1$. Then, $\Gamma(1) = 1$, and so Equation (12) can be rewritten as:

$$i(t) = \mu \frac{\Gamma(q)}{\Gamma(1+q)} (1-t_s)^{q-1} + \varphi. \quad (16)$$

Because $\Gamma(1+q) = q\Gamma(q)$, this can be expressed as follows:

$$i(t) = \frac{\mu}{q} (1-t_s)^{q-1} + \varphi, \quad (17)$$

when $q > 1$, Equation (17) is a monotonically increasing function, and when $0 < q < 1$, Equation (17) is a descending function and $i(t) > 0$. If $q < 0$, then Equation (17) is a descending function and $i(t) < 0$.

In other situations, we define $q = 1$ so that $\Gamma(1) = 1$ and $(1-t_s)^q - 1 = 1$. Then, Equation (12) can be rewritten as follows:

$$i(t) = \mu \frac{\Gamma(p)}{\Gamma(p+1)} t_s^{p-1} + \varphi. \quad (18)$$

As above, $\Gamma(p+1) = p\Gamma(p)$, so this can be expressed as:

$$i(t) = \frac{\mu}{p} t_s^{p-1} + \varphi. \quad (19)$$

Equation (19) is similar to Kostikov's infiltration model, and the coefficient μ/p is equivalent to Kostikov's model coefficient (Kostikov, 1932; Parhi, Mishra, & Singh, 2007), that is, $\mu/p = an$. When $0 < p < 1$, Equation (19) is a monotonically decreasing function, and when $p > 1$, Equation (19) is a monotonically increasing function; when $p = 1$, Equation (19) is a constant φ .

For $p = 0.5$, Equation (19) can be rewritten as follows:

$$i(t) = 2\mu t^{-0.5} + \varphi. \quad (20)$$

This is similar to Philip's water infiltration model. The difference is that the coefficient in Philip's model is 0.5S (Philip, 1957), whereas the coefficient of Equation (20) is 2μ , which is also a monotonically decreasing function.

From the above, we can assume that Horton's infiltration model is a special case of Equation (10), whereas the Kostikov and Philip infiltration models are special cases of Equation (20). The mathematical GMF and BF models have an extremely wide range of applications.

5 | CONCLUSION

Unlike HS, in which the CI increases monotonously with time and the IR decreases monotonously, W-RS exhibits the following characteristics: (a) A two-stage CI with an overall growth phenomenon and an IR with a mutation phenomenon. Larger values of ISWC produce an earlier inflection point in the CI and larger values of IR at the inflection point. (b) IR has a single peak when the initial infiltration process (stage I) is neglected, and the stable post-peak IR is higher than the pre-peak value.

The applicability of KF, PKF, GF, PGF, FSF, GMF, BF, and PBF models was analysed using HS and W-RS samples. The KF, GMF, and BF functions were found to be suitable for HS. For W-RS, the GMF function not only reflects the monotonous decrease in infiltration ($s \leq 1$), but also recreates the complete infiltration process, namely the gradual decline in IR in the initial stage, gradual increase before the inflection point, subsequent gradual decrease after the inflection point, and final stable infiltration ($s > 1$). The BF model reflects the monotonous decrease process in HS ($0 < p \leq 1$ and $q > 1$). PBF gives a U-shaped IR that decreases gradually from the initial point and then gradually increases before the inflection point, and also reflects the IR with a right-skewed distribution curve about the left and right inflection points. Therefore, the BF/PBF model offers the better simulation accuracy and has the widest applicability.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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